

Coordinated Action in Multiteam Systems

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This study investigated coordinated action in multiteam systems employing 233 correspondent systems, comprising 3 highly specialized 6-person teams, that were engaged in an exercise that was simultaneously “laboratory-like” and “field-like.” It enriches multiteam system theory through the combination of theoretical perspectives from the team and the large organization literatures, underscores the differential impact of large size and modular organization by specialization, and demonstrates that conventional wisdom regarding effective coordination in traditional teams and large organizations does not always transfer to multiteam systems. We empirically show that coordination enacted across team boundaries at the component team level can be detrimental to performance and that coordinated actions enacted by component team boundary spanners and system leadership positively impact system performance only when these actions are centered around the component team most critical to addressing the demands of the task environment.

Keywords: multiteam systems, differentiated team roles, team of teams, dynamic centrality, coordination

Teams have emerged as the basic building blocks of organizations, and interest in team-based research has grown in response to this shift (Chen, Kanfer, DeShon, Mathieu, & Kozlowski, 2009; LePine, Piccolo, Jackson, Mathieu, & Saul, 2008). More complex task environments often necessitate the formation of large teams, especially when a broad set of highly specialized skills is required. Increases in information processing and coordination demands can become quite problematic, however, as group size and task specialization increase (Galbraith, 1977; Gooding & Wagner, 1985; Steiner, 1972). In fact, Hackman (2002) concluded, “When group size becomes very large, the problems generated far outweigh the incremental resources brought by the additional members” (p. 117).

One solution to the problems induced by large size and task specialization is to adopt a team-of-teams, or multiteam system organizational form. Mathieu, Marks, and Zaccaro (2001) defined a multiteam system as a collective of two or more interdependent teams organized as a tightly coupled activity system (Thompson, 1967). Each team in the system possesses specialized skills, capabilities, and functions that uniquely contribute to achieving a shared, superordinate goal. It is this shared, superordinate goal that defines the boundary of the multiteam system, and a multiteam system can reside within a single legal entity (e.g., one organization) or multiple entities (e.g., an alliance of organizations).

Multiteam systems are a hybrid organizational form, part traditional team and part large organization. Like a traditional team, members of a multiteam system share responsibility for outcomes, the task environment is conducive to teamwork and requires members to work together interdependently, there are clear membership boundaries, and authority to manage the system’s work processes is clearly specified (Guzzo & Dickson, 1996; Hackman, 1987, 2002). Yet unlike a traditional team, multiteam systems are too large and specialized to effectively employ direct mutual adjustment among each and every member of the system. Like modular organizations (e.g., cross-functional and multiple business, or “M-form,” organizations; Chandler, 1962; Denison, Hart, & Kahn, 1996; Helfat & Eisenhardt, 2004), work within a multiteam system is compartmentalized and undertaken by specialized subunits (component teams) that focus on particular tasks and work collaboratively to address the needs of the task environment. Yet unlike these large organizational forms, multiteam systems are not overly constrained by standardization, institutionalized procedures, or formal rules. This latter point is exceedingly important because multiteam systems face highly turbulent environments that require rapid response to changing circumstances (Mathieu et al., 2001).

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Mathieu et al. (2001) argued that “we understand very little about the influence of teams on other teams, as well as how multiple teams coact in pursuit of common goals” (p. 292). Although their work goes a long way toward establishing a solid theoretical foundation for multiteam system research, opportunities still exist to strengthen the explanatory power of the theory. This is particularly true with respect to the complexity of interactions within multiteam systems; these require more sophisticated boundary management and coordination mechanisms than do traditional stand-alone teams (Marks, DeChurch, Mathieu, Panzer, & Alonso, 2005). Extant multiteam system theory is silent on what these mechanisms are and how they operate, however. The hybrid nature of multiteam systems suggests that a synthesis of micro and macro perspectives would aid our understanding of coordination processes and mechanisms in these systems and inform theory development.

Multiteam systems reside in a performance environment that is not conducive to either rigorous laboratory or extensive field study (Mathieu, Maynard, Taylor, Gilson, & Ruddy, 2007). As a consequence, scant research has focused on the multiteam system organizational form despite its importance in many applied settings. When researchers have turned their attention to multiteam systems, the tendency has been to study small collectives (4–6 members) organized into component teams with limited unique specialization. Although this research is informative, theory in the areas of workflow design and large organizations (e.g., Sinha & Van de Ven, 2005; Thompson, 1967) leads us to question whether these studies trigger important intra- and inter-team dynamics that occur in multiteam systems. To this point, theories of workflow design in large organizations suggest that unrestricted, direct coordination (i.e., allowing everyone to interact with everyone else) will be detrimental to multiteam system performance, yet prior research (e.g., Marks et al., 2005) has reported only positive relationships between both within- and cross-team processes and multiteam system performance.

This study contributes to the literature in three ways. First, theoretically, we enhance the explanatory power of multiteam system theory by challenging the generalizability of theoretical conclusions from small traditional teams to larger multiteam systems. Combining perspectives from the team literature (e.g., Hackman, 1987, 2002; Humphrey, Morgeson, & Mannor, 2009; Kozlowski & Bell, 2003) and the large organization literature (e.g., Astley & Zajac, 1990, 1991; Caplow, 1957; Davis, Eisenhardt, & Bingham, 2009; Sherman & Keller, 2011), we develop a theory-informed perspective on coordination in multiteam systems. We argue that direct mutual adjustment among all members in the collective, a cornerstone to the effectiveness of small teams, is actually detrimental to performance in multiteam systems. Effective management of information processing complexity, resulting from large size and modular organization by specialization, precludes direct mutual adjustment among all members in the multiteam system and makes a formal structure for boundary spanning and coordination a necessity.

The enhanced theory suggests that coordination enacted by different actors in a multiteam system impacts performance differentially and that these systems require coordination via mechanisms that can leverage differences in relative importance resulting from degree of centrality to the mission and structural position in the system. It also establishes the concept of differentiated

component team roles as a decision heuristic that provides an efficacious means of prioritizing conflicting coordination demands by designating the component team most critical to addressing the needs of the task environment as the nexus of the system.

Second, empirically, we tested multiteam system theory in a context that was simultaneously “laboratory-like” and “field-like” by employing multiteam systems comprising multiple highly specialized component teams (three 6-person teams). This context extends the external validity of multiteam system theory. Further, we develop and test hypotheses regarding horizontal coordinated action (i.e., coordination that takes place between peer actors within an organizational level) within and among component teams and vertical coordinated action (i.e., coordination that takes place between report-to and direct reporting actors across organizational levels whose breadth of scope and authority differ) between component teams and a team formally tasked with coordination and system integration. Reported results provide initial support for the enhanced theory, indicating that mechanisms universally thought to improve performance in teams negatively impact performance in multiteam systems, and illustrate the positive performance impact resulting from the enactment of differentiated component team roles as a coordinating mechanism.

Last, practically speaking, this study provides valuable insights to managers and leaders tasked with designing, implementing, and maximizing the performance of these complex organizations.

Theory and Hypothesis Development

A multiteam system is a hybrid organizational form that is especially well suited for addressing the demands of today’s complex and highly turbulent environments. Component teams composed of highly task specialized resources are the primary interface with the task environment and sit at the lowest organizational level in a multiteam system (Mathieu et al., 2001). Each component team is reciprocally interdependent with at least one other component team in the system, and, thus, the key to coordination at this level is effective mutual adjustment (Thompson, 1967). Those aspects of coordination that are beyond the scope of the component teams are handled by a team situated hierarchically above them (Thompson, 1967). With the best overall picture of activity within the system (Martin & Eisenhardt, 2010) and the broadest understanding of capabilities and strategy, the primary function of the team at this level is integrative leadership (Sherman & Keller, 2011). The integration team comprises boundary spanning representatives from each component team and roles with systemwide responsibilities (e.g., leadership).

Large size and modular organization by specialization are fundamental attributes of multiteam systems that make an enormous difference in the nature of coordinated action. Relational complexity, the division of labor into differentiated team roles and the need for hierarchy and formal boundary spanning mechanisms to facilitate coordination, is a direct result of increases in size and specialization. The hypotheses that follow address the differential performance impact of horizontal coordinated action among the members of a highly specialized component team situated at the forefront of the multiteam system–task environment interface (the “point” team) and the members of other highly specialized component teams providing complementary expertise in support of the superordinate mission (“support” teams), horizontal coordi-

nated action among the boundary spanners for the component teams, vertical coordinated action between the members of each component team and their respective boundary spanner, and vertical coordinated action between boundary spanners and multiteam system leadership (see Figure 1).

Coordinated Action Enacted at the Component Team Level

Coordinated action among individuals, the orchestrated sequencing and timing of interdependent actions, is a key behavioral process that affects performance in teams (Kozlowski & Bell, 2003; Marks, Mathieu, & Zaccaro, 2001). Because multiteam systems are organized as a team of individual component teams, there is every reason to expect that coordinated action is a key behavioral process that affects performance in multiteam systems as well. Each component team within a multiteam system is conceptually a traditional team, and therefore the positive relationship between intrateam coordination and performance reported in the team literature (Ilgen, Hollenbeck, Johnson, & Jundt, 2005; Kozlowski & Ilgen, 2006) should hold within each component team (e.g., Marks et al., 2005).

Interdependence of action is an attribute of both teams (Kozlowski & Bell, 2003) and multiteam systems (Mathieu et al., 2001); however, multiteam systems are larger entities than traditional teams (Mathieu et al., 2001). Larger organizations group homogenous positions together (i.e., departmentalize) in order to maximize the benefits of interdependence and minimize the costs of coordination (Thompson, 1967). As described by Sinha and Van de Ven (2005, p. 393), "The basic design principle is to group together the most highly interdependent activities within units first, then the next-most interdependent interfaces among units reporting to a common supervisor, and the least interdependent activities among different departments and functions (Thompson, 1967)."

Grouping priority is given to the technology or task environment dimension where the benefits derived from coordination are the greatest.

Multiteam systems follow this design principle, known as the principle of hierarchical decomposition (Sinha & Van de Ven, 2005). The most interdependent roles are grouped together first to form component teams, and then, because each component team shares input, process, and outcome interdependencies with at least one other component team, they are grouped together to form a multiteam system. Thus, the benefits of coordination that accrue within component teams outweigh the benefits of coordination across component team boundaries. Therefore, we posit

Hypothesis 1: Horizontal coordinated action among members of the same component team has a greater impact on multiteam system performance than does horizontal coordinated action among members of different component teams.

Multiteam systems address highly turbulent environments that place a premium on their ability to respond rapidly to changing circumstances; thus, some component teams process interdependencies are necessarily reciprocal and intense (Mathieu et al., 2001). Uncertainty increases as interdependence becomes more reciprocal (Gresov, 1989; Tushman, 1978, 1979), resulting in an increase in requirements for information exchange and processing among interdependent units (Galbraith, 1977; Gittell, 2002; Gresov, 1989; Ito & Peterson, 1986; Keller, 1994; Tushman & Nadler, 1978). The importance of information processing to processes of coordination and the impact of coordination on performance also increase as interunit interdependence increases (Sherman & Keller, 2011).

Reciprocal and intense interdependence requires coordination by mutual adjustment; however, mutual adjustment is most demanding of communication and decision effort (Thompson, 1967).

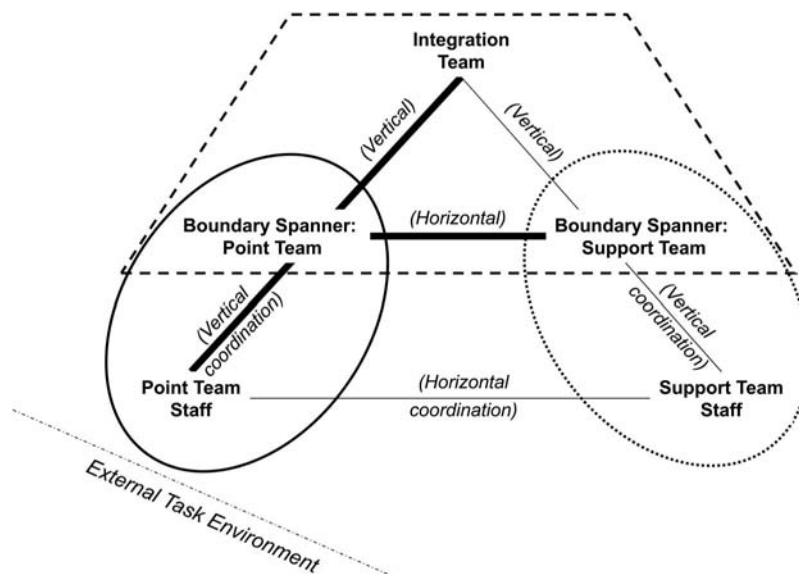


Figure 1. Depiction of horizontal and vertical coordination. Solid lines depict coordinating relationships between focal actors. Bold lines represent hypothesized positive relationships between coordinated action and performance.

Unfortunately, the number of potential relationships within a collective grows exponentially as membership increases (Caplow, 1957), leading to additional information processing complexity. Large organizations manage the detrimental effects of information processing complexity and minimize the costs of coordination resulting from interunit interdependence, size of membership in the collective, and departmentalization by handling those aspects of coordination that are outside the scope of any individual unit through hierarchy (Sinha & Van de Ven, 2005; Thompson, 1967).

A multiteam system is larger than a traditional team. It is departmentalized into highly specialized component teams, and the teams in the system exhibit reciprocal and sometimes intense process interdependence. This suggests a degree of information processing complexity that is analogous to that in a large organization. Collectively, these arguments suggest that those aspects of coordination which are outside the scope of any individual component team should be handled through hierarchy and that direct coordination among members of different component teams in a multiteam system is counterproductive. Thus, in contrast to findings previously reported in the literature (e.g., Marks et al., 2005), we propose

Hypothesis 2: Horizontal coordinated action among the staff members of different components teams exhibits a negative relationship with multiteam system performance.

Coordinated Action Enacted at the System Level

Interdependencies and specialization create a clear need for effective coordination across component team boundaries (Ancona & Caldwell, 1992), yet large size and information processing requirements argue for the restriction of direct interaction among each and every person within a multiteam system. Further, grouping individuals into highly specialized component teams (i.e., departmentalization) increases the possibility of strong group cohesiveness and identification within these teams, intergroup competition, and a singular focus on in-group goals at the expense of superordinate goals (Li & Hambrick, 2005; Tajfel, 1982). In-group bias can lead to negative stereotyping of other groups, which becomes a justification for maintaining social distance and secrecy (Ashforth & Mael, 1989). Even in contexts where open competition does not occur, there is still the potential for each specialized, cohesive, and finely tuned subteam to develop separate and disjointed perspectives and mental models (Cannon-Bowers, Salas, & Converse, 1993). This alone could disrupt the ability to directly coordinate activities across subteams. Ironically, then, factors that contribute to effective processes within teams can lead to coordination difficulties among teams.

Richter, West, Van Dick, and Dawson (2006) proposed that problems arising between groups can often be solved by members with dual attachments. Their proposal aligns with the perspective espoused by scholars in the large organization tradition. These scholars argue that effective coordination requires the creation of integrative liaison and leadership roles as interdependence becomes more reciprocal (Burns, 1989; Keller, 1994; Sherman & Keller, 2011). Building on the work of Thompson (1967), Galbraith (1977) proposed a graduated sequence of integration modes whereby each particular mode is efficacious over a limited range of interdependence. Low degrees of interdependence can be coor-

ordinated effectively via standard operating procedures and organizational hierarchy. However, according to Sherman and Keller (2011), "When the volume of direct contact between two interdependent units increases with multiple personnel communicating, coordination problems can develop if a common point of contact does not exist in each unit" (p. 248). The component teams in a multiteam system are highly interdependent; thus, theory and research lead to the conclusion that a formal boundary spanning mechanism is necessary for effective coordination in these systems.

Introducing formal mechanisms for coordination into a multiteam system, whether integrative liaison and/or leadership roles, places responsibility for coordination among component teams into the hands of a small group of boundary spanners. This reestablishes the potential for mutual adjustment because the number of boundary spanning communication links is reduced substantially. It does not, however, alter the fact that "work systems located within and between organizations operate in contexts of multiple and often conflicting contingencies" (Sinha & Van de Ven, 2005, p. 397). For example, decisions taken to improve interunit coordination may preclude a component team boundary spanning representative from being well coordinated simultaneously with their own subteam and the other boundary spanners. This leads us to conclude that multiteam systems do require sophisticated boundary management processes, as suggested by Marks et al. (2005). We suggest that insight into these processes lies in the concept of functional interdependence.

Functional interdependence. Interaction complexity and conflicting contingencies are resultant attributes of task interdependence in networked systems of work, and increases in subunit specialization lead to greater interdependencies among the subunits of an organization (March & Simon, 1958; Sinha & Van de Ven, 2005). Further, functional interdependence is characteristic of organizational systems where the activities of subunits are highly specialized and goals are collectively shared (Astley & Zajac, 1991; Holm, Eriksson, & Johanson, 1999). As described by Gresov and Drazin (1997), "function can be defined as the way in which a component part of a subsystem (i.e., a structure) contributes to the maintenance of the system and its ability to be adaptive to its environment. Function refers to the ability of the system to maintain interdependence with other social actors or environments" (p. 406).

A subunit's importance is directly related to the extent to which the functions it performs are central and critical to the flow of work (Astley & Zajac, 1990; Hickson, Hinings, Lee, Schneck, & Penning, 1971; Olsen, 1978). Although both are attributes of an organization's resources and input-output interdependencies, centrality is prototypically associated with position in the flow of work, the structure of an organization, or the flow of information, whereas criticality is an attribute that indicates the degree to which a subunit is irreplaceable. Thus, theory and research on the division of labor suggest that there will be a team (or teams) within a functionally interdependent system of work that has greater influence and impact on collective performance relative to other teams because it occupies a position in the workflow that is more central and/or performs specialized functions that are more critical.

In addition, Humphrey et al. (2009) argued and demonstrated "that certain team roles are most important for team performance and that the characteristics of the role holders in the 'core' of the

team are more important for overall performance" (p. 48). They suggest three criteria that define the strategic core of a team: roles that "(a) encounter more of the problems that need to be overcome in the team, (b) have a greater exposure to the tasks that the team is performing, and (c) are more central to the workflow of the team" (Humphrey et al., 2009, p. 50). Recent research regarding the strategic core of a team also suggests that a functionally equivalent and homologous phenomenon exists at the team-of-teams level of analysis (Chen, Bliese, & Mathieu, 2005; Gresov & Drazin, 1997).

Differentiated component team roles. Within a multiteam system, labor is divided into specialized and functionally interdependent component teams to most effectively address the demands of the external task environment. As just discussed, theory suggests that one or more of the component teams will be at the fulcrum of activity—most central and/or most critical—and that coordination within and around these teams will be of primary importance (Astley & Zajac, 1991; Hickson et al., 1971; Humphrey et al., 2009). We refer to these as *point teams*. The other highly specialized and functionally unique component teams act in the capacity of *support teams*, providing information and expertise beyond that represented in the point team. On the basis of this theoretical concept of differentiated component team roles, we posit

Hypothesis 3a: Vertical coordinated action between the point team boundary spanner and point team staff members exhibits a positive relationship with multiteam system performance.

Hypothesis 3b: Vertical coordinated action within the point team (i.e., between the point team boundary spanner and the staff members) has a greater impact on multiteam system performance than does vertical coordinated action within the support team (i.e., between the support team boundary spanner and the staff members).

As discussed previously, the research of Richter et al. (2006) and of scholars in the large organization tradition (e.g., Sherman & Keller, 2011) suggests that integrative liaison and/or leadership roles are necessary for effective coordination in a multiteam system. Further, as Thompson (1967) articulated, "Each level [in a hierarchy] is not simply higher than the one below, but is a more inclusive *clustering* or combination of interdependent groups, to handle those aspects of coordination which are beyond the scope of any of its components" (p. 59). As the purpose of each integrative level is to handle only that which cannot be handled at lower levels in the system, a high degree of activity at this level is indicative of unresolved issues or inefficiencies in the system: for instance, ineffective workflow design (Sinha & Van de Ven, 2005), a lack of understanding of the team interaction model (Zaccaro, Rittman, & Marks, 2001), or social loafing (Gooding & Wagner, 1985; Steiner, 1972) on the part of component team staff members, or some combination of these and/or other factors.

We argued previously that problems associated with multiple collaborations (Sinha & Van de Ven, 2005) may preclude a component team boundary spanner from being well coordinated simultaneously with his or her own subteam and the other boundary spanners. We argued further that a consequence of component team specialization is that one team, the point team, is most central

and critical to meeting the demands of the task environment at any given point in a performance episode. A de facto result of centrality and criticality is that coordination around the point team becomes of utmost importance. This suggests that support team boundary spanners should focus on being well coordinated with the point team boundary spanner even if this makes them appear uncoordinated with their own team.

Collectively, these arguments suggest that performance will be high whenever vertical coordinated action within the point team (between point team boundary spanner and his or her staff) is high, regardless of the degree of vertical coordinated action within the support team (between support team boundary spanner and his or her staff). They also suggest that performance will be highest whenever vertical coordination is high in all component teams. Finally, these arguments suggest that multiteam system performance will be low whenever coordinated action within the point team is low and coordinated action within the support team is high (e.g., when coordination is centered on the support team instead of the point team and in instances where point team boundary spanners are poorly coordinated with their staff and boundary spanners representing support teams fail to follow suit). Thus, we propose

Hypothesis 4: Vertical coordinated action within the support team will interact with vertical coordinated action within the point team such that multiteam system performance will be highest when vertical coordinated action within both component teams is high and lowest when vertical coordinated action within the support team is high and vertical coordinated action within the point team is low.

Complex organizations such as cross-functional teams and multiteam systems, where subunits are networked together from a variety of organizational sources, require an integrative leadership role (Sherman & Keller, 2011). Information processing and complexity theories (Anderson, 1999; Helfat & Eisenhardt, 2004) argue that leaders overseeing the complete network of modular subunits are in the best position to orchestrate the most effective cross-unit collaborations because they are most likely to have the best "big picture" information (Martin & Eisenhardt, 2010). Once again, this does not mean that a high degree of activity at the integration level is desirable; the role of integrative leadership is to handle those aspects of coordination that are beyond the scope of levels lower in the hierarchy (Thompson, 1967). However, these arguments do suggest that when leaders do engage in integrative activities, coordination around the component team enacting the role of point team is of primary importance. Thus, we propose

Hypothesis 5: Vertical coordinated action between the boundary spanner for the point team and leadership exhibits a positive relationship with multiteam system performance.

Martin and Eisenhardt (2010) recently challenged the dominant "corporate-centric process" perspective, derived from information processing and complexity theories (Anderson, 1999; Helfat & Eisenhardt, 2004), which argues that leaders atop a network of modular subunits are best positioned to direct cross-unit collaborations. These scholars argue that subunit leaders sit at the collaboration decision nexus and thus are most likely to have the most relevant, big-picture understanding of cross-unit collaborations

and the most intimate knowledge of the operating details necessary to accomplish them. Indeed, Martin and Eisenhardt (2010) showed empirically that processes enacted at the subunit coordination level lead to better cross-unit collaborations than do more corporate-centric processes. This suggests that coordination activities collectively undertaken by the component team boundary spanners should have a greater impact on performance than those undertaken by multiteam system leadership. Therefore,

Hypothesis 6a: Vertical coordinated action undertaken by component team boundary spanners, as a group, has a greater impact on multiteam system performance than does vertical coordinated action undertaken by leadership.

As discussed in detail in the arguments leading up to Hypothesis 4, higher levels in the hierarchy of a multiteam system exist primarily to “handle those aspects of coordination which are beyond the scope of any of its components” (Thompson, 1967, p. 59). Further, in line with the hierarchical decomposition principle of workflow design (Sinha & Van de Ven 2005; Thompson, 1967), the most highly interdependent activities are grouped together at the lowest level in the hierarchy, and the next-most interdependent interfaces are grouped together at the next highest level in the hierarchy (e.g., under a common manager). This suggests that coordination activities at the component team level should have a greater impact on multiteam system performance than coordination activities at higher levels in the system. Therefore, we posit

Hypothesis 6b: Coordinated action undertaken by members of the component teams has a greater impact on multiteam system performance than does coordinated action undertaken at all levels higher in the system.

Method

Sample and Data Source

The sample consisted of 3,262 U.S. Air Force captains with 5–9 years of experience in a broad range of professional disciplines. These officers were attending a 5-week leadership course offered periodically at an Air Force base located in the southeastern United States. The participants were assigned to 233 fourteen-person teams using criteria designed to ensure that team composition was equivalent on factors including experience, job function, gender, and age. Team assignments were kept intact throughout the 5-week course; thus, these teams had a meaningful past and future. Participants were evaluated throughout the course, and these ratings became a part of the package that went to their promotion board. Hence, participants were experienced professionals motivated to perform well on this exercise.

Overview of the Experimental Protocol

Task environment. The Leadership Development Simulator (LDS), jointly developed by the U.S. Air Force and a large research university, was specifically designed to present a complex multiteam system task environment in which individuals and teams must collaborate to effectively manage a large number of resources under time pressure. Teams of up to 14 people coordinate activities and integrate multiple sources of information with the common objective of finding and engaging targets located in a predefined but active environment. The environment is enacted in a grid, 16 rows (1–16) by 16 columns (A–P), totaling 256 cells (depicted in Figure 2). LDS is programmed to capture objective

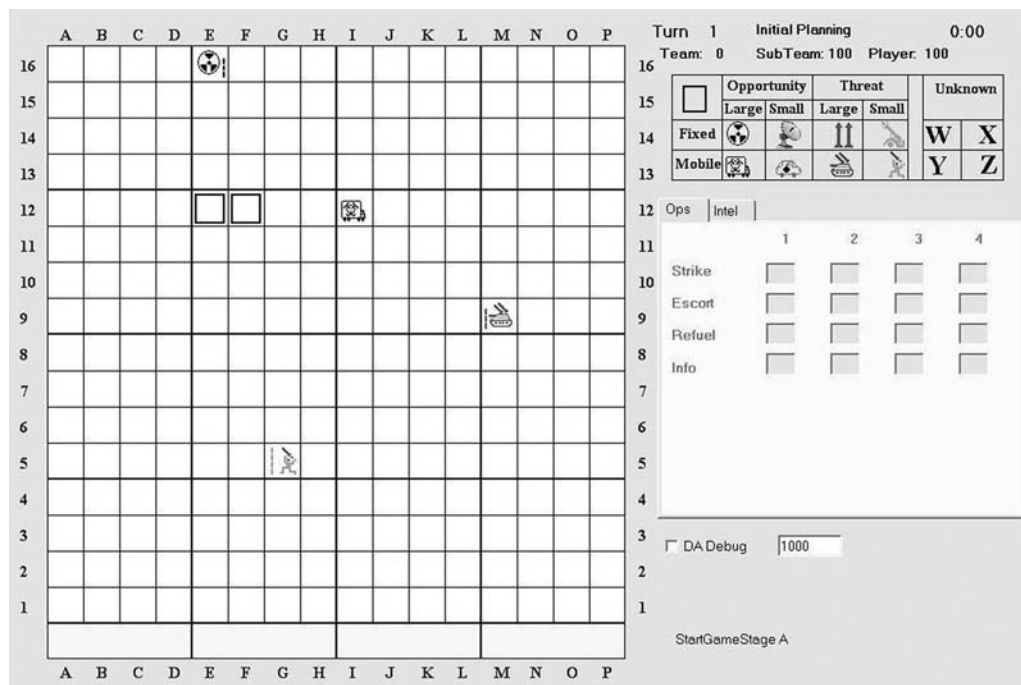


Figure 2. Task environment illustrated by a sample Common Operational Picture (COP).

measures associated with individual behaviors, coordinated action, and performance. The simulator created a highly controlled *laboratory-like* context, where each and every multiteam system engaged in the same task, with the same resources, time limits, training, and personnel types. It was also *field-like* in that activities related to tasks that participants were likely to perform, support, or lead in their professional lives.

Multiteam system objective. The superordinate objective of the multiteam system was to maximize points scored across a single performance episode. Performance was impacted by four types of events: Points were gained with the destruction of an opportunity or a threat, and points were lost when a remotely piloted aircraft (RPA) was destroyed or a base was attacked. Teams were awarded the most points for a large opportunity, fewer points for a small opportunity or a large threat, and the least points for a small threat. Destroyed assets were replenished at the start of each round to ensure consistency of resources across rounds; however, targets were permanently removed from the environment once they were destroyed.

Performance episode. A performance episode unfolded in a series of action and transition phases similar to those described by Marks et al. (2001) and lasted approximately 2 hours. This was followed by a 30-min feedback session conducted by a subject matter expert. Teams were given an initial intelligence briefing at the start of the performance episode and were allowed 10 minutes to formulate a strategy (i.e., a transition phase) prior to engaging in the 10-round preprogrammed scenario. Each round consisted of a sub-episode during which assets were deployed and missions enacted (i.e., activities fitting Marks et al.'s definitions of coordination, team monitoring, and backup) and a sub-episode during which feedback was received and the environmental situation analyzed (i.e., activities fitting Marks et al.'s definitions of system monitoring and monitoring progress toward goals). Control of and decision-making authority over assets passed from component team staff members to boundary spanners to multiteam system leadership during the first sub-episode of a round.

During the second sub-episode of each round, team members received feedback detailing the results of their actions, which the team then used to develop a shared understanding of the task environment and a plan for the next round. Team members worked collaboratively to construct and maintain the Common Operational Picture (COP), a graphic representation of the task environment as they collectively perceived it (i.e., location of threats and opportunities). All members of the multiteam system could readily view the COP and were responsible for ensuring that it was an accurate reflection of the situation. (Figure 2 is an illustration of the COP.)

At the start of the performance episode, teams were presented with an unknown task environment. Hidden throughout the 256 cells of the task environment were various targets of differing characteristics. "Opportunities" gained the team points when effectively engaged, whereas "threats" would attack the team's assets and cost the team points if not properly engaged. Some targets were large, some small; some were stationary, others were mobile; and some required coordinated effort to be engaged while others did not. Point values varied across targets, and targets were permanently removed from the environment once they were destroyed. The performance episode reported here occurred at the beginning of the second week of the course, a day after the training.

Training. All 233 multiteam systems received the same comprehensive training. This included prereading material, a 45-min illustrated slide presentation, and 15 minutes of hands-on training and directed practice using a scenario specifically designed for training purposes. All participants completed a performance episode on the same day as the training, and hence they were highly proficient at the technical aspects of the task.

Multiteam System Structure and Team Roles

Roles were arranged in a multiteam system structure consisting of two six-member component teams and a six-member integration team (see Figure 3). The integration team had decision authority

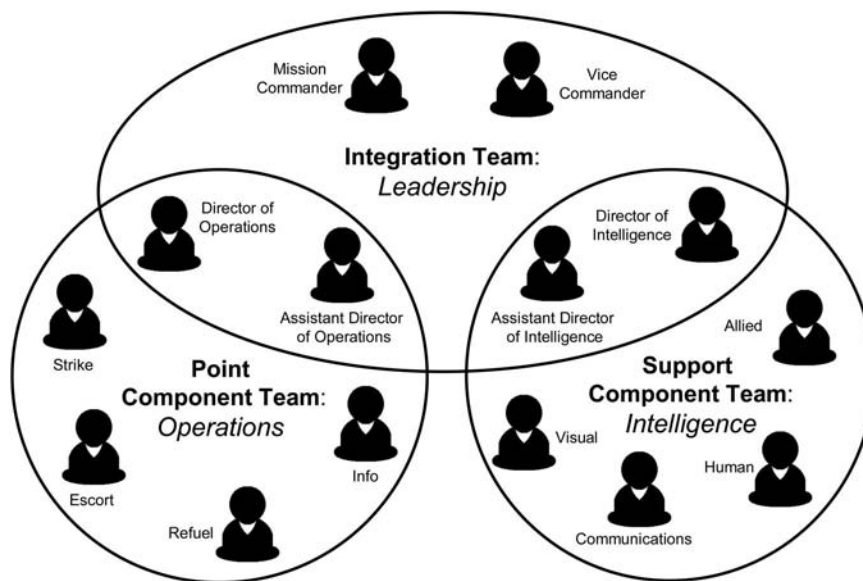


Figure 3. Conceptualization of the multiteam system structure employed.

over the component teams and thus enacted leadership. There were no physical barriers to discussion (all members of the multiteam system were co-located). Each role was narrowly defined, with specific responsibilities and objective measures of performance.

Component teams. Each component team consisted of four staff members in charge of a unique set of nontransferable assets, along with a director and an assistant director. The Operations team reflected the point team in this context. Members of the Operations team had control of all RPAs and were tasked with engaging targets while avoiding destruction. RPAs were specialized in that only one (Strike) could successfully engage opportunities, only one (Escort) could successfully engage threats, one (Refuel) was necessary for other RPAs to reach distant portions of the environment, and one (Info or Recon) had only the capability to collect information about the task environment from three locations at once. All other RPAs and all intelligence assets controlled by the support team collected information from the location to which they were deployed only. A single RPA could successfully engage a small target, whereas two RPAs were needed to engage a large target. Each member of the point team controlled four assets of a single type (e.g., the Strike role controlled the four Strike RPAs) for a total of 16 Operations assets.

The component team in the support role, operationalized as the Intelligence team, gathered supplemental information about the environment for the Operations team. This team controlled assets capable of gathering information, probabilistically, regarding whether there was an opportunity or threat in a single location. Representative of their support team role, intelligence assets only passively observed the environment and could not destroy targets nor could they be destroyed. There were four different types of intelligence assets (Visual, Communications, Allied, and Human), and each support team member controlled eight assets of a single type (e.g., the Visual role controlled the eight Visual assets) for a total of 32 intelligence assets.

Each intelligence asset type was 95% accurate in half of the task environment (South, North, Central, or Borders) and gave misleading (95% inaccurate) information, both false positives and false negatives, when deployed to the other half of the task environment (see Figure 4). Importantly, the specific area of the task environment where each type of intelligence asset was accurate (its “sweet spot”) was unknown at the start of the simulation and thus had to be learned via collective trial-and-error experience. This was a “connect the dots” type of problem-solving task (Ellis et al., 2003) where it was nearly impossible for a single team member to quickly discover the sweet spot of his or her asset type

based on experience alone. Multiteam system members had to collaborate on actions and share their experiences to discern where each intelligence asset type was accurate. Discovering this early was critically important to the success of the multiteam system. Without knowledge of each intelligence asset’s sweet spot teams did not know the accuracy of the information they were uncovering, and thus the team’s understanding of the changing task environment could be confounded by bad intelligence.

Integration team. The integration team construct was operationalized as a leadership team in this study. The six-member Leadership team comprised a mission commander, vice commander, and the directors and assistant directors from the two component teams. The Leadership team was responsible for planning, monitoring, and managing disagreements between component teams, and this team had final decision-making authority on the deployment of all assets. The directors and assistant directors possessed dual memberships in both a component team and the Leadership team and thus provided a formal means for boundary spanning.

Half of the Leadership team had ultimate responsibility for ensuring that the multiteam system’s COP was accurate and up to date. The vice commander was responsible for building and updating the COP, and the assistant directors assisted this effort, acting as information conduits from their respective component teams. Importantly, no single member of the multiteam system could acquire or process all of the information, and thus it took the collaborative effort of all members to build the COP. Whereas the vice commander and the two assistant directors were focused on synthesizing information from past actions, the other half of the Leadership team (the mission commander and the two component team directors) was focused on future actions. This half of the Leadership team had the authority to modify any deployment decision made by team members lower in the multiteam system structure and was tasked with ensuring coordination. The directors oversaw their respective teams, and the mission commander oversaw the complete system. Leaders faced a large number of decisions, information search complexities, and a small decision window. Time simply did not allow higher levels in the structure to question or change every decision made at lower levels.

Measures

The simulator (LDS) was programmed to capture objective measures of key behaviors associated with coordination and performance. Thus, the central constructs under investigation were

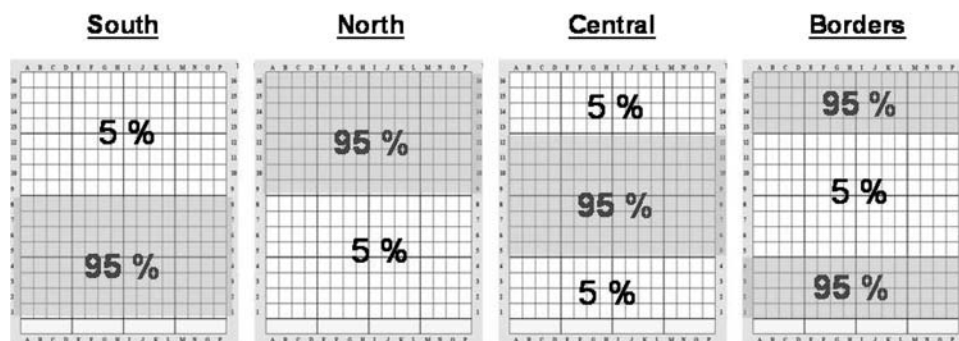


Figure 4. Intelligence asset sweet spot (95% accurate/5% inaccurate) zones.

measured via objective indices rather than retrospective self-reports.¹ Many forms of coordination took place, and we measured seven different types that specifically represent mutual adjustment. These were (a) horizontal coordination within the four-person point team, (b) horizontal coordination within the four-person support team, (c) horizontal coordination between the component teams, (d) vertical coordination between members of the point team and its boundary spanner, (e) vertical coordination between members of the support team and its boundary spanner, (f) vertical coordination between the point team boundary spanner and multiteam system leadership, and (g) vertical coordination between the support team boundary spanner and multiteam system leadership.

Horizontal coordinated action behavioral indicators. Horizontal coordination within the four-person point team was operationalized in terms of the frequency with which two or more members deployed their respective (and task-unique) assets (RPAs) to the same location during the same round. Given that there were 256 locations and only four assets that were controlled by each member of the point team, the odds that any such joint mission would happen by chance was remote. Thus, this behavior was an indicator that two members of the point team were explicitly coordinating their actions.

As we noted earlier, identifying which intelligence asset type was accurate in each sweet spot zone was a connect-the-dots type of problem-solving task (Ellis et al., 2003), and no one individual could determine this efficiently based solely on his or her own experience. Members of the support team had to synchronize their queries, pool their experiences, and share results to learn which intelligence asset type was accurate in each of the four zones. Further, teams that worked together effectively to solve this problem quickly were able to deploy intelligence assets more efficaciously. For example, a team that solved this problem early could coordinate and thus maximize the use of intelligence assets by deploying each asset to a different location where accuracy was high in subsequent rounds, whereas a team that never solved this problem would not be able to maximize the use of intelligence assets in any round. Because the odds of both solving the connect-the-dots problem and maximizing the use of intelligence assets by chance in the absence of coordinated action was very low, we operationalized horizontal coordinated action within the four-person support team as the frequency that team members deployed intelligence assets to cells where accuracy was high.

Horizontal coordinated action between the component teams was measured as the frequency with which members of the point and support team deployed their assets into the same cell during a round. Again, given the large number of cells and the small number of assets, the odds that any such joint mission would happen by chance was remote; thus, the frequency was an indicator that members of the two different component teams were coordinating their actions.

Vertical coordinated action behavioral indicators. As discussed previously, decisions made at one level in the multiteam system structure could be reviewed, modified, or accepted unchanged at the next level. Because effective coordination across levels removed any requirement to modify decisions at a higher level, vertical coordination was operationalized as the frequency with which decisions made at one level were modified by the next level, reverse coded so that higher values represented greater coordination (i.e., less modification).

Multiteam system performance. Performance was operationalized as the sum of points gained (from targets destroyed) minus points lost (from RPAs destroyed and bases attacked) during the 10-round performance episode.

Results

Table 1 presents descriptive statistics and correlations for all study variables. Tables 2, 3, and 4 contain the tests of our hypotheses, which were directional and hence tested via one-tailed tests. We entered the control for performance during training in Step 1 of all regressions; it accounted for roughly four percent of the variance explained.

Coordinated Action Within and Between Component Teams

Component team members. Hypothesis 1 posited that coordinated action enacted by component team members within the same team has a greater impact on multiteam system performance than does coordinated action enacted by component team members from different teams. We used proportion of variance effect size, calculated as Cohen's f^2 , to test this hypothesis (Cohen, 1988). By convention, f^2 values of .02, .15, and .35 are respectively considered to represent small, medium, and large effect sizes. As shown in Table 2, horizontal coordinated action within the component teams has a medium and statistically significant effect on performance ($f^2 = .22$), whereas horizontal coordinated action between component teams has a small ($f^2 = .02$) and statistically insignificant effect size, supporting Hypothesis 1.

The effects of horizontal coordinated action within and between component teams were tested by regressing multiteam system performance on the three component-team-level coordinated action variables. The results of this analysis can be found in Table 3, Step 2. Note that, as expected, the second and third rows of this table show that horizontal coordinated action within both component teams exhibit significant positive relationships with overall multiteam system performance (point: $\beta = .22, p < .05$; support: $\beta = .34, p < .05$). Thus, high levels of direct coordinated action within the component teams resulted in higher levels of multiteam system performance. This is consistent with results from the traditional team literature.

The fourth row of Table 3 shows that horizontal coordinated action between component teams exhibits a significant *negative relationship* with performance ($\beta = -.12, p < .05$). This is consistent with Hypothesis 2, which posited that unbridled, direct coordinated activity between multiteam system members across the point-support team boundary would be counterproductive. Support for Hypothesis 2 challenges the notion that coordination

¹ To check construct validity, we compared our objective measure of vertical coordination to a subjective measure of coordination perceptions based on the two coordination effectiveness items from Rapp and Mathieu (2007). Because our access to the professionals involved in this study was limited, a convenience sample of MBA students (17 teams of five) engaged in a similar LDS exercise was used. Across 34 performance episodes these measures were positively related at $r = .43$ when adjusted for the reliability of the scale ($r_{xx} = .77$). This provides support for the validity of our measures (Schwab, 1980).

Table 1
Descriptive Statistics and Correlations

Variable	<i>M</i>	<i>SD</i>	1	2	3	4	5	6	7	8
1. Performance during training	59.86	51.97	—							
2. Horizontal coordinated action within point component team	97.86	20.05	.11	—						
3. Horizontal coordinated action within support component team	204.91	40.57	.10	.08	—					
4. Horizontal coordinated action between component teams (point and support)	41.48	19.93	-.13*	.00	.05	—				
5. Vertical coordinated action between point team staff and boundary spanning representative	-23.57	10.54	-.11	.46*	-.06	.01	—			
6. Vertical coordinated action between support team staff and boundary spanning representative	-43.27	21.52	-.03	-.08	.49*	.06	-.10	—		
7. Vertical coordinated action between point team boundary spanner and leadership	-11.61	7.03	-.05	.30*	.00	.05	.35*	-.10	—	
8. Vertical coordinated action between support team boundary spanner and leadership	-13.39	11.66	-.02	-.10	.32*	.03	-.05	.49*	-.08	—
9. Multiteam system performance	149.38	47.20	.20*	.26*	.37*	-.11	.15*	-.05	.19*	-.07

Note. $N = 233$. 14-member multiteam systems.

* $p < .05$, two-tailed.

among team members is always good and contradicts previously reported findings from both the traditional literature on teams and the multiteam systems literature (e.g., Marks et al., 2005). This result indicates that large size and modular organization by specialization, two attributes that are definitional for multiteam systems (Mathieu et al., 2001), induce dynamics that differ from those found in traditional teams.

Component team boundary spanners. Hypothesis 3a stated that coordinated action between members of the point team and their boundary spanner will be positively related to performance. The results of this analysis can be found in Table 4, Step 2. As is

apparent in Row 2, the predicted relationship holds true for the point team, supporting this hypothesis ($\beta = .17, p < .05$). This suggests that a multiteam system requires good coordination between the point team and its respective boundary spanner. Note from Row 3 that there was no parallel finding for the support team ($\beta = -.02, ns$). Thus, a lack of coordination between the support team and its boundary spanner was not necessarily detrimental to performance. This may reflect a decision on the part of the support team boundary spanner to be well coordinated with the point team boundary spanner at the expense of being well coordinated with his or her staff.

Table 2
Proportion of Variance Effect Sizes for Coordinated Actions

Variable	Component team			Leadership			Multiteam system		
	R^2		Cohen's f^2	R^2		Cohen's f^2	R^2		Cohen's f^2
Horizontal coordinated action within point component team	0.206*	0.219*	0.22				0.219*		0.28
Horizontal coordinated action within support component team									
Horizontal coordinated action between component teams (point and support)									
Vertical coordinated action between point team staff and boundary spanning representative	0.068*	0.069*	0.03	0.081*		0.02	0.101*	0.297*	0.11
Vertical coordinated action between support team staff and boundary spanning representative	0.040*		0.00						
Vertical coordinated action between point team staff and boundary spanning representative									
Vertical coordinated action between support team staff and boundary spanning representative									
Vertical coordinated action between point team boundary spanner and leadership				0.079*		0.02			
Vertical coordinated action between support team boundary spanner and leadership									

Note. $N = 233$. Performance during training included as a covariate in all R^2 s (per Cohen, 1988).

* $p < .05$, two-tailed.

Table 3

Component Team Level: Effects of Horizontal Coordinated Action Within and Between Teams

Variable	Multiteam system performance			
	β		R^2	ΔR^2
	Step 1	Step 2		
Step 1				
Performance during training	.20*	.12*	.038*	.038*
Step 2				
Horizontal coordinated action within point team		.22*	.219*	.181*
Horizontal coordinated action within support team		.34*		
Horizontal coordinated action between component teams		-.12*		

Note. $N = 233$.* $p < .05$, one-tailed.

We employed Cohen's f^2 as a measure of effect size to test Hypothesis 3b (Cohen, 1988). This hypothesis posited that coordinated action between members of the point team and their boundary spanner would have a greater effect on performance than would coordinated action between members of the support team and their boundary spanner. The results reported in Table 1 show that coordinated action between members of the point team and their boundary spanner have a small but statistically significant effect on performance ($f^2 = .03$), whereas the effect of coordinated action between members of the support team and their boundary spanner was negligible ($f^2 = .00$). Thus, Hypothesis 3b was supported.

Hypothesis 4 investigated the combined effects of coordination enacted by the boundary spanning representatives of the component teams. It posited that multiteam system performance would be highest in systems where boundary spanners are all well coordi-

nated with their respective staff members and lowest in those systems where the point team boundary spanner was highly uncoordinated with the point team staff (i.e., made a lot of adjustments) while the support team boundary spanner remained well coordinated with the support team staff.

To test this hypothesis, we added the interaction between (a) vertical coordinated action between the point team and its boundary spanning representative and (b) vertical coordinated action between the support team and its boundary spanning representative to the regression reported in Table 4. As indicated in Step 3, the relationship between this variable and multiteam system performance is significant and in the direction predicted ($\beta = .11$, $p < .05$), indicating support for Hypothesis 4. A plot of this interaction (see Figure 5) shows that performance is high when the point team boundary spanner is well coordinated with the team staff, irrespective of degree of

Table 4

Leadership (Integration) Team Level: Effects of Coordinated Action Within and Between Leaders

Variable	Multiteam system performance				R^2	ΔR^2
	β					
	Step 1	Step 2	Step 3	Step 4		
Step 1						
Performance during training	.20*	.21*	.20*	.20*	.038*	.038*
Step 2						
Vertical coordinated action between point team staff and boundary spanning representative		.17*	.16*	.12*	.069*	.030*
Vertical coordinated action between support team staff and boundary spanning representative		−.02	−.04	−.01		
Step 3						
Vertical coordinated action between point team staff and boundary spanning representative X			.11*	.10	.081*	.012*
Vertical coordinated action between support team staff and boundary spanning representative						
Step 4						
Vertical coordinated action between point team boundary spanner and leadership				.14*	.101*	.020*
Vertical coordinated action between support team boundary spanner and leadership				−.05		

Note. $N = 233$.* $p < .05$, one-tailed.

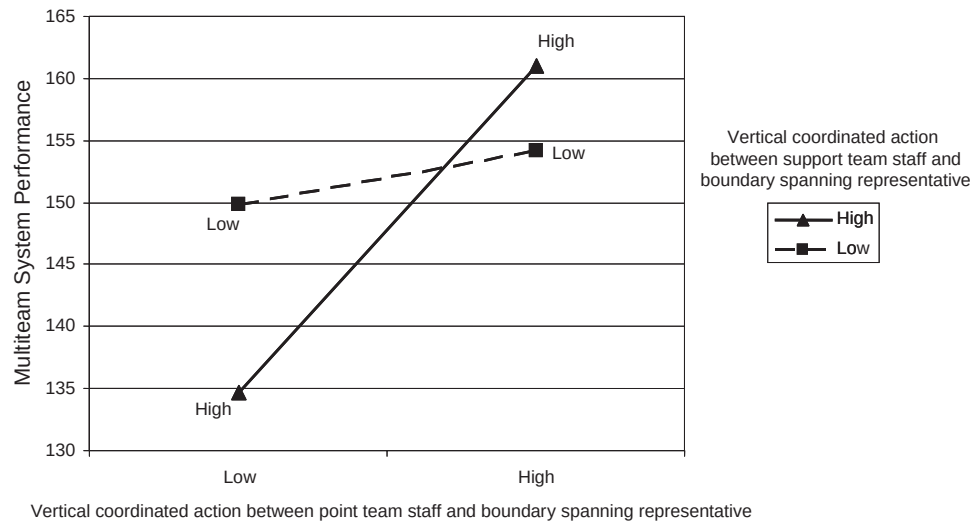


Figure 5. Interaction between vertical coordinated action at the component team level (“High”/“Low” represents ± 1 standard deviation).

vertical coordination within the support team, and highest when the boundary spanners for the point and support teams are both well coordinated with their staff.

The plot also indicates that the worst case scenario is when the boundary spanner for the point team is poorly coordinated with his or her team yet the boundary spanner for the support team remains highly coordinated with his or her staff. As the plot illustrates, when the boundary spanner for the point team is highly uncoordinated with the point team staff, it is important for the support team boundary spanner to be in “lockstep” (i.e., be uncoordinated with the support team staff as well). Although not definitive, this implies that multiteam system performance suffers when the boundary spanner for the support team pays more attention to being well coordinated with his or her staff than to being well coordinated with the boundary spanner for the point team.

We conducted a simple slopes analysis (Aiken & West, 1991) as a further test of Hypothesis 4 and found the slope to be positive and significant when vertical coordinated action between the support team and its boundary spanning representative was high ($\beta = .19$, $p < .05$, at 1 SD and $\beta = .01$, ns , at -1 SD , two-tailed). Crossover occurred at vertical coordinated action between the point team and its boundary spanning representative equal to 0.39 of a standard deviation, well within the observed range. These results are consistent with and support Hypothesis 4.

Coordinated Action Across Multiteam System Levels

Hypothesis 5 posited that vertical coordinated action between the boundary spanner for the point team and leadership exhibits a positive relationship with multiteam system performance. The results of this analysis (see Table 4, Step 4) indicate that this predicted relationship holds true for the point team ($\beta = .14$, $p < .05$). In combination with Hypothesis 3a, this suggests that a multiteam system needs to be well coordinated vertically across all point team boundaries. Note also that there was no parallel finding

for the support team ($\beta = -.05$, ns) suggesting, once again, that lack of vertical coordination within the support team does not affect performance.

The last two hypotheses concerned the performance impact of coordination decisions taken by roles at different levels in the system. The results are reported in Table 2. Hypothesis 6a focused on coordination enacted by component team boundary spanners versus leadership. It posited that, collectively, component team boundary spanners will have a greater impact on performance than will leaders. Employing Cohen’s f^2 as a measure of effect size, we found that, inconsistent with Hypothesis 6a, vertical coordinated action undertaken by the component team boundary spanners, as a group, and the vertical coordinated action of leadership both have a small and statistically insignificant effect on performance ($f^2 = .02$ and $f^2 = .02$, respectively).

Hypothesis 6b focused on coordination enacted at the component team level versus that which occurs higher up in the hierarchy, positing that coordinated action undertaken by the members of the component teams has a greater impact on multiteam system performance than does action undertaken at all levels higher in the system. Employing Cohen’s f^2 as a measure of effect size, we found support for the hypothesized relationship. Coordinated action undertaken at the level of the component teams has a statistically significant, medium effect on performance ($f^2 = .28$), whereas the effect of coordinated action undertaken at all levels higher in the system was small ($f^2 = .11$), although statistically significant.

Finally, we combined all coordination variables employed in this study into a single regression analysis and found that parameter estimates remained quite similar (see Table 5). Vertical coordinated action between the point team staff and their boundary spanning representative turned nonsignificant, however, when the measures of coordination at the component team level were entered into the regression (Step 5), while vertical coordinated action between the support team staff and their

Table 5

Effects of All Coordinated Action Variables Within and Between Teams (Tables 3 and 4 Combined)

Variable	Multiteam system performance					R ²	ΔR ²
	β						
	Step 1	Step 2	Step 3	Step 4	Step 5		
Step 1							
Performance on prior scenario	.20*	.21*	.20*	.20*	.12*	.038*	.038*
Step 2							
Vertical coordinated action between point team staff and boundary spanning representative		.17*	.16*	.12*	.07	.069*	.030*
Vertical coordinated action between support team staff and boundary spanning representative		−.02	−.04	−.01	−.21*		
Step 3							
Vertical coordinated action between point team staff and boundary spanning representative X			.11*	.10	.07	.081*	.012*
Vertical coordinated action between support team staff and boundary spanning representative							
Step 4							
Vertical coordinated action between point team boundary spanner and leadership				.14*	.10*	.101*	.020*
Vertical coordinated action between support team boundary spanner and leadership				−.05	−.09		
Step 5							
Horizontal coordinated action within point team					.11*	.297*	.196*
Horizontal coordinated action within support team					.48*		
Horizontal coordinated action between component teams					−.11*		

Note. $N = 233$.* $p < .05$, one-tailed.

boundary spanning representative turned significant. Both of these effects reflect partialing issues resulting from the high degree of correlation between the measure for vertical coordinated action between component team staff members and their boundary spanning representative and the measure for horizontal coordinated action within the focal component team (point team: $r = .46$, $p < .05$; support team: $r = .49$, $p < .05$). Taken as a whole, the set of coordination measures explained 30% of the variance in multiteam system performance.

Discussion

Many of today's most challenging task contexts require the resource capabilities of a large organization but the mutual adjustment agility of a traditional small team. Examples include incident command systems (Bigley & Roberts, 2001), emergent groups responding to disasters (Majchrzak, Jarvenpaa, & Hollingshead, 2007), and organizations facing a market crisis (e.g., Apple and the iPhone 4 antenna issue). Combining these perspectives, we enrich multiteam system theory.

Theoretical Contribution and Empirical Results

We challenged the generalizability of theoretical conclusions emanating from the traditional team literature, arguing that unstructured and unbridled, direct mutual adjustment, a cornerstone to the effectiveness of small teams, is actually detrimental to performance in multiteam systems. We found that while coordinated action within component teams was positively related to multiteam system performance, direct coordinated activity among

all members of the point and support teams actually harms multiteam system performance. Both results were as expected, although the latter result conflicts with previous research that employed collectives of just two or three two-person subteams. Small collectives are indistinguishable from traditional teams and thus do not exhibit all the dynamics found in multiteam systems. Our findings with larger 14-person multiteam systems demonstrate that fundamental attributes of multiteam systems—large size and subteam specialization—alter the dynamics within these systems and that failure to utilize a formal coordination structure causes the system to revert into a large and undifferentiated team, with predictably negative results.

We also argued that coordination enacted by different actors in a multiteam system impacts performance differentially, and this is what we found. As expected, the effect of coordination on performance is greater for the actions taken by the component team members than for actions taken by members of the leadership team ($f^2 = .27$ vs. $f^2 = .11$). Component team members sit at the multiteam system–task environment boundary and thus are best positioned to effectively address the uncertain and shifting demands of the external environment. Leaders, on the other hand, have less intimate familiarity and up-to-the-moment knowledge of the environment. They also have a broader set of responsibilities to attend to in a limited amount of time and thus are tasked with only those things that cannot be handled at the component team level (Sinha & Van de Ven, 2005; Thompson, 1967). Although leadership oversight is important, component team members enact the most critical roles during periods of action.

We did not find support for our prediction that boundary spanners would have a greater effect on performance than would leaders higher up in the hierarchy. It is very possible that this effect was constrained by the fact that this study constituted only one performance episode (i.e., it could be that differences at higher levels only exhibit across multiple performance episodes). It also is interesting to note that the positive effect on performance of coordinated action taken among members of the same component team outweighs the negative effect of coordinated action among members of different component teams ($f^2 = .21$ vs. $f^2 = .02$). Based on the theoretical design principle of hierarchical decomposition, this finding is actually not at all surprising.

A central property of a multiteam system is that the subdivision of work into specialized task activities at the component team level reduces the requirement for direct contact among all members of the system. Component teams are smaller and thus, as in stand-alone teams, complexity is manageable and accountability is promoted. Unlike stand-alone teams but like large organizations, activities across the multiteam system can be coordinated because component teams are linked by a team whose primary responsibility is boundary spanning and coordination. Moreover, the size of the team tasked with coordination is small enough to support mutual adjustment. Thus unlike large organizations but like stand-alone teams, the multiteam system as a whole can be flexible and react to a dynamic task environment despite its relatively large size.

This simple compare-and-contrast discussion highlights a fundamentally important, yet overlooked, attribute of multiteam systems—their hybrid nature—and leads to the logical conclusion that a full understanding of the dynamics of coordination in these systems requires the synthesis of traditional team and large organization theoretical perspectives. We offer the following cursory description of these complex dynamics. Large size and modular organization by specialization lead to problematic information processing complexity that precludes direct mutual adjustment among all members of the system (it is not only untenable, it is actually detrimental to performance). In addition, coordination enacted by different actors in the system impacts performance differentially. This is a consequence of differences in relative importance associated with a focal actor derived from degree of centrality to the mission and structural position in the system.

Last, effective management of this inherent information processing complexity gives rise to the need for a formalized boundary spanning and coordination structure and makes mechanisms that can leverage differences in relative importance, such as differentiated component team roles (i.e., point, support, and integration), a necessity. Although previous multiteam system theorizing has not conceptually differentiated subteam roles, we argued that a multiteam system is composed of two different types of component teams—point and support—linked via boundary spanners with dual membership in a component team and the integration (leadership or liaison) team. We support this conception drawing on extant organizational theory (Astley & Zajac, 1990, 1991; Delery & Shaw, 2001; Humphrey et al., 2009; Pearsall & Ellis, 2006).

This study is unique in that component team boundary spanners were faced with active decision makers above, below, and lateral to them, all of whom might demand accommodation. Because of the complexities of this multiple collaboration problem (Sinha &

Van de Ven, 2005), it was easy for individuals in these positions to look “uncoordinated” with one or more of these other parties. By noting the different and specialized nature of the point, support, and integration teams, however, we were able to show that total coordination is not a requirement for high multiteam system performance. Rather, the key is selective coordination through the boundary spanner for the point team and addressing the needs of this core team (Humphrey et al., 2009).

As the results of this study indicate, multiteam system performance is highest when the boundary spanner for the point team is well coordinated with point team staff members, the boundary spanner for the support team, and multiteam system leadership. Lack of coordination between the boundary spanner for the support team and his or her team, or multiteam system leadership, did not have detrimental effects on multiteam system performance. The analysis reported in Table 5 infers a complementary conclusion; coordination centered on the support team, rather than the point team, is detrimental to performance. The apparent lack of vertical coordination in the support team may reflect a price worth paying in order to accommodate the needs of the point team, an insight that cannot be derived from extant multiteam system theory because it does not distinguish between the different types of specialized subteams that constitute a multiteam system. Thus, this study extends multiteam system theory by establishing differentiated team roles as a mechanism that enables effective mutual adjustment in multiteam systems.

Managerial Implications

These findings have two critically important implications for the managers tasked with leading multiteam systems. First, they show that teams that enact differentiated team roles as a mechanism to achieve coordination consistently outperform teams that act like one large undifferentiated team (i.e., everyone interacting with everyone). This recommendation for large multiteam systems stands in contrast to the prescription to simply open up all the communication channels among team members, as is implied by research on small stand-alone teams. Second, boundary spanning managers are often “stuck in the middle” and unable to be well coordinated with all the parties that might demand accommodation. The distinction between component team roles (point vs. support) provides a useful heuristic for prioritizing irreconcilable coordination demands. Importantly, the recommendation to give priority to the demands of the point team during action phases runs counter to more traditional perspectives that advocate either reliance on the formal authority structure (i.e., hierarchy alone) or empowerment of each subteam to take decisions independently of the other subteams.

Limitations and Future Directions

Several limitations of our study can be noted to help guide future research. First, although coordination by mutual adjustment was the construct of interest in this study, it does not encompass all forms of coordination in these systems. Further, although frequency of real-time, coordinated behavior is a valid measure of this construct, it provides only an indirect indication of process quality. Process quality can also be measured subjectively through the self-reported ratings of participants or observers. This simula-

tion was effective at documenting coordination and performance, but due to the field setting we were not in a good position to systematically observe competitive behaviors or conflict, even though these are central aspects of many theories of intergroup behavior and teams. The organizational setting and the realities of running 25–35 fourteen-person multiteam systems simultaneously precluded direct observation, videotaping, or extensive surveying of all the teams. Thus, there is a need for additional research using these and other measures to validate the findings reported here.

Second, our measures of vertical coordination were operationalized as the frequency with which decisions made at one level were modified by the next level, reverse coded so that higher values represented greater coordination (i.e., less modification). We believe this measure accurately portrays the intended construct, but it is possible that the measure is capturing alternative constructs, such as the upward exercise of voice, processes related to conflict, leadership styles, and/or other factors that differ from coordination. Thus, future research should seek to clarify the validity of objective measures of vertical coordination, such as the ones employed here, as well as study vertical coordination using alternative measurement approaches.

Third, our study is premised on the supposition that large size and modular organization by specialization induce important dynamics that affect performance. Although the findings we reported point in this direction, future research is needed to more fully explore the implications of these contingencies with other variables known to be important. For example, this study employed a functional multiteam system structure in which the point and support teams did qualitatively different work. A multiteam system could also be structured divisionally, whereby the subteams target the unique needs of different geographic regions or product markets and must share resources to accomplish their objectives. The dynamics of a multiteam system with this structure might be drastically different. Such a structure makes it easier to understand and appreciate what the other subteams are doing, but it also makes it easier to see the other team as a direct competitor for limited resources, setting the stage for conflict.

Fourth, we theoretically developed the concept of differentiated team roles and showed that both coordination and system performance are enhanced as suggested by this conceptualization. This provides some initial support; however, it is hardly conclusive. In addition, multiteam systems face highly dynamic environments that require rapid response to changing circumstances (Mathieu et al., 2001). Shifts in environmental demands can alter goal priorities and task importance, leading to changes in the degree of centrality of teams in the flow of work. A focal component team might be central and/or critical at one moment and less so at another or vice versa. This suggests that multiteam systems must be effective at shifting the differentiated roles enacted by component teams. The literature has yet to consider this phenomenon of “dynamic centrality,” despite the potential implications for goal pursuit and performance.

Further, our theoretical development stipulates that the role enacted by the integration team varies along a continuum between “leadership” and “liaison,” yet all the integration teams in the study reported here were empowered with decision authority and thus were representative of the leadership end of the continuum only. Research is needed to investigate the effects of liaison teams tasked with spanning component team boundaries and mediating

the relationships among them without the power of formal authority. In short, additional research is needed to more fully test the validity of the differentiated team role conceptualization, its implications, and the assertion that it is widely applicable to the multiteam system organizational form.

Last, the current study examined multiteam systems from a macro- or system-level perspective. Future research might benefit from a more microlevel approach. For instance, the disposition of individuals in key roles (e.g., boundary spanners) may influence behavior in and performance of multiteam systems. In addition, our study focused on one complete performance episode, but many important factors affecting performance and other outcomes of interest can be understood only at the multi-episode, sub-episode, and process levels of analysis. Decisions are extremely path dependent, for example. Thus, future research should also investigate effects that emerge over time.

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Correction to Messersmith et al. (2011)

In the article “Unlocking the Black Box: Exploring the Link Between High-Performance Work Systems and Performance” by Jake G. Messersmith, Pankaj C. Patel, and David P. Lepak (*Journal of Applied Psychology*, 2011, Vol. 96, No. 6, pp. 1105–1118), some information concerning the data collection process was omitted from the original published version. All online versions have now been corrected. The details of the corrections are as follows:

Julian Gould-Williams has been added as the fourth author on the paper.

The following citation has been added to the reference list: Gould-Williams, J. (2006–2008). *Wales Local Government Staff Survey, 2006–2008* [computer file]. Colchester, Essex: UK Data Archive [distributor], August 2009. SN: 6239, <http://dx.doi.org/10.5255/UKDA-SN-6239-1>

The following text has been added to the author note: We acknowledge the Economic and Social Research Council, and the UK Data Archive for data access and funding. The original data creator, depositor, or copyright holders, the funder of the data collection, and the UK Data Archive bear no responsibility for the analysis or interpretation of the data.

The corrected version of the article is available here: <http://dx.doi.org/10.1037/a0024710>

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